

2 Plan design & data: Assignment

Instructions

Submit your answers to the seven first exercises through Aula Global. The remainder of your assignments concern self-studies.

Describing a study

Quick reminder from class: Kanwal et al. (2017) taught subjects an artificial language. This language only has three (made up) words: *zopudon*, *zopekil* and *zop*. The first two words referred to distinct objects (think: *zopudon* means apples and *zopekil* means bananas). The short form *zop* could mean either (think: fruit). That is, *zop* is ambiguous between the two meanings, and thus can lead to misunderstandings. They had subjects communicate about the two meanings (apples/bananas) in pairs, alternating who was the speaker and who was the receiver. The core motivation for this study was to see if people would use the ambiguous word, *zop*, even though it is more risky than the unequivocal but longer alternatives *zopudon* and *zopekil* under certain experimental manipulations.

Here's a glimpse of the data of the speakers:

```
df <- read.csv('https://tinyurl.com/2s3p9s2z')
head(df)
```

##	pairnum	IP	trial	display	label
## 1	1	67.85.42.18	1	0	zop
## 2	1	67.85.42.18	2	3	zopudon
## 3	1	67.85.42.18	3	0	zop
## 4	1	67.85.42.18	4	0	zopekil
## 5	1	67.85.42.18	5	2	zopudon
## 6	1	67.85.42.18	6	1	zopekil

The column **pairnum** identifies each pair of subjects (alternating speaker and hearer roles); the **IP** column identifies each subject; **trial**s keep track of the order in which the turns happened (trial 1 is the first time a subject is a speaker, trial 2 is the second, and so on); **display** codes whether one type of object (0/1) or another type of object (2/3) was displayed to the speaker; and **label** shows what the speaker actually said to communicate this object.

1. What kind of study is this? Observational, experimental, or simulation?
2. What kind of variable is **pairnum**?
3. What kind of variable is **trial**?
4. What kind of variable is **label**?

Inspect the sender data from Kanwal et al. (2017) yourself.

5. How many times did the sender with IP 172.91.192.214 say *zop*?
6. How many pairs of subjects participated in the experiment?
7. How many sender trials did the experiment have for each subject?

Self-study

1. Work through Chapter 4 and 6 of *Introduction to Data Analysis* (Franke 2021). If you need a refresher on descriptive statistics, also look at Chapter 5;
2. Answer these four questions concerning your analysis report:
 - What is your general research question (e.g., How does a learner’s first language affect learning a second language?)
 - Why do you think this question is interesting? What does an answer to it tell us?
 - What is your specific research question (e.g., Is there a difference between the speed in which Spanish and German monolinguals learn Catalan?)
 - What kind of data would you use to address Question 2 if you had unlimited resources?
 - What kind of data are you planning to use to address 2 within the scope of this class?
 - How will you obtain it?
 - How much will you collect?
 - Do you think that is enough data to address Question 2? Why (not)?

Elaborate with as much detail as you (currently) have but it is perfectly acceptable if you answer in bullet points and single sentences. You will be asked to submit these answers for next week’s assignment. Randomly selected peers will give you feedback on them.

References

- Franke, Michael. 2021. *An Introduction to Data Analysis*.
- Kanwal, Jasmeen, Kenny Smith, Jennifer Culbertson, and Simon Kirby. 2017. “Zipf’s Law of Abbreviation and the Principle of Least Effort: Language Users Optimise a Miniature Lexicon for Efficient Communication.” *Cognition* 165: 45–52. <https://doi.org/10.1016/j.cognition.2017.05.001>.